IE 7374 · Generative AI | Industry Hackathon

HACKATHON PROBLEM STATEMENT

Open-Data Intelligence for Distributed Solar Development

Industry Partner: PowerTrust | Spring 2026

01 BACKGROUND & OVERVIEW

The global push toward renewable energy has made distributed solar a critical component of future energy systems. However, understanding what it actually takes to build and operate a distributed solar farm in a given country remains highly complex and fragmented. Developers and investors must navigate a wide range of factors — economic, regulatory, infrastructural, and socio-political — often buried across disparate and unstructured public data sources.

This is a real-world problem that companies like PowerTrust tackle regularly when establishing a foothold in a new country or adapting to evolving laws and regulations in the renewable energy sector. Currently, much of this research is conducted manually by team members, making it time-intensive and difficult to scale. This project represents an important opportunity to automate and standardize these workflows, reducing costs and improving speed as operations expand.

02 CORE CHALLENGE

Central Question

Design and prototype a data-driven system or model that answers: What does it take to successfully build a distributed solar farm in a specified country? (e.g., India, Mexico, Malaysia). The solution should leverage publicly available data to construct a holistic, structured, and interpretable view of the solar development landscape.

Why Build a Model Instead of a Static Report?

A model enables dynamic, repeatable, and scalable analysis. It allows users to:

- Answer follow-up questions without restarting the analysis
- Generate evolving insights as new data becomes available
- Extend the framework to new regions or countries with minimal rework
- Interact with the model through a conversational chat interface, asking natural language questions about any dimension without re-running the full pipeline

As part of your solution, you will present a case study for a chosen country, including state- or province-level variation where applicable. You should also clearly outline your methodology for updating the model over time, incorporating new data sources, insights, and expanding to additional geographies.

You may find that companies such as Halcyon.io have developed AI-powered platforms to achieve similar outcomes, primarily focused on the U.S. market. In contrast, this challenge emphasises applications in the Global South, where data availability, structure, and regulatory landscapes can differ significantly. You are encouraged to draw inspiration from such platforms, but your solution should reflect original thinking in how your model addresses the problem and extracts insights.

03 KEY QUESTIONS TO ADDRESS

Your model must extract, synthesise, and present insights across the following six dimensions:

1. Cost & Economics

Estimate and/or benchmark:

- CAPEX and O&M costs for distributed solar
- Utility and grid charges
- Comparative cost structures across generation types (e.g., small/large solar, wind, CCGT, gas)

2. Grid Access & Queue Dynamics

- Average grid interconnection wait times
- Regional variation in grid congestion or accessibility

3. Subsidies, Incentives & Policy Support

- Net metering policies
- Solar-specific subsidies, grants, or tax incentives

4. Utility Standards & Obligations

- Renewable Portfolio Standards (RPS) or equivalent mandates
- Utility obligations to procure renewable energy

5. Public Comment & Approval Signals

- Key themes, objections, or concerns raised during project approval processes
- Insights derived even from unstructured or incomplete records

6. Unknown Unknowns — Discovery Layer

Surface country-specific risks or constraints not explicitly defined above. Examples may include:

- Energy access gaps

- Land-use conflicts
- Financing barriers
- Regulatory bottlenecks

04 EXPECTED MODEL OUTPUTS

Your solution must go beyond data collection and produce a clear, decision-oriented output, including:

- **Structured Report or Dashboard** Summarising each of the six dimensions above
- **Visualisations** Cost breakdowns, grid wait times by region, policy coverage maps, etc.
- **Scoring or Insight Layer** Helps interpret feasibility or risk for a chosen market
- **Conversational Chat Interface** Allows users to query the model in natural language without re-running the full pipeline

Conversational Chat Interface

The chat interface is a core deliverable, not an optional add-on. It must be grounded in the data your model has already retrieved and analysed — not a general-purpose LLM response. The interface must support all of the following interaction types:

- **Dimension queries** *e.g., "What is the average CAPEX for a 1 MW rooftop solar installation in Tamil Nadu?"*
- **Comparison queries** *e.g., "How do grid interconnection wait times in Maharashtra compare to Karnataka?"*
- **Risk and gap queries** *e.g., "What data is missing that would most affect our CAPEX estimate?"*
- **Multi-turn follow-ups** *The interface must maintain conversational context so that follow-up questions build on prior answers within the same session*

Important constraint: Responses must be grounded in your retrieved data. When the model cannot answer from available data, it must say so explicitly and identify what additional data would be needed. Hallucinated or fabricated answers will be penalised during evaluation.

Data Availability Audit

Explicitly identify all of the following in your submission:

- What data was successfully sourced and used
- What critical data is missing or not publicly available
- How these gaps impact decision-making

05 DELIVERABLES

Working Prototype	Model / system with full source code submitted to GitHub, including the conversational chat interface
Case Study	Applied to one country: India, Malaysia, Mexico, or Brazil — including state/province-level variation where available
Documentation	Data sources, assumptions, methodology for updating the model, and data availability audit
Presentation	Concise slide deck presenting findings, insights, and a live demo

06 EXAMPLE PUBLIC DATA SOURCES

> *The sources below are illustrative examples for India. You are expected to identify equivalent sources for your chosen country (India, Malaysia, Mexico, or Brazil).*

1. Cost & Economics — CAPEX, O&M, Tariffs, Cross-Technology Comparison

- **Central Electricity Regulatory Commission** https://cercind.gov.in
  - **Documents:** *Tariff Orders, Benchmark Capital Cost Reports*
- **Solar Energy Corporation of India** https://www.seci.co.in
  - **Documents:** *Auction Results, Power Purchase Agreements (PPAs)*
- **Ministry of New and Renewable Energy** https://mnre.gov.in
  - **Documents:** *Rooftop Solar Programme Guidelines, Benchmark Costs*
- **International Renewable Energy Agency (IRENA)** https://www.irena.org
  - **Reports:** *Renewable Power Generation Costs*
- **Lazard** https://www.lazard.com
  - **Reports:** *Levelized Cost of Energy (LCOE)*

2. Grid Access & Queue Dynamics

- **Central Electricity Authority** https://cea.nic.in
  - **Documents:** *National Electricity Plan, Transmission Plans*
- **POSOCO** https://posoco.in
  - **Reports:** *Congestion Reports, Real-Time Grid Data*
- **Power Grid Corporation of India** https://www.powergrid.in
  - **Documents:** *Transmission Project Reports*
- **Maharashtra Electricity Regulatory Commission** https://www.merc.gov.in
  - **Documents:** *Interconnection regulations and filings*
- **Open Access Registry India** https://openaccessregistry.com
  - **Data:** *Open access approvals and transaction records*

3. Subsidies, Incentives & Policy Support

- **Ministry of New and Renewable Energy** https://mnre.gov.in
  - **Documents:** *Rooftop Solar Programme Phase II*
- **Central Electricity Authority** https://cea.nic.in
  - **Documents:** *Model Net Metering Frameworks*
- **NITI Aayog** https://www.niti.gov.in
  - **Reports:** *Energy policy and subsidy frameworks*

4. Utility Standards & Obligations

- **Central Electricity Regulatory Commission** https://cercind.gov.in
  - **Documents:** *Renewable Purchase Obligation (RPO) Regulations*
- **Karnataka Electricity Regulatory Commission** https://kerc.karnataka.gov.in
  - **Documents:** *State RPO targets*
- **Central Electricity Authority** https://cea.nic.in
  - **Reports:** *Renewable integration studies*
- **Ministry of Power India** https://powermin.gov.in
  - **Documents:** *National Electricity Policy*

5. Public Comment & Approval Signals

- **Ministry of Environment, Forest and Climate Change** https://moef.gov.in
  - **Documents:** *EIA Reports, Public Hearing Transcripts*
- **Maharashtra Pollution Control Board** https://mpcb.gov.in
  - **Documents:** *Public consultation records*
- **National Green Tribunal** https://greentribunal.gov.in
  - **Documents:** *Case filings and rulings*
- **Press Information Bureau** https://pib.gov.in
  - **Releases:** *Project announcements*
- **PRS Legislative Research** https://prsindia.org
  - **Summaries:** *Policy debates and stakeholder inputs*

Good luck — build something that would actually work in the field.